

**The Impact of the Real Exchange Rate on Economic Growth: A Comparative Study of
OECD and Non-OECD Countries**

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Abstract

This study examines the relationship between the real exchange rate (RER) and economic growth in OECD and non-OECD countries using panel regression and time-series analyses. The panel estimates indicate a U-shaped relationship for OECD countries and an inverted U-shaped relationship for non-OECD countries. Complementary vector autoregression (VAR) analyses of China and Japan provide country-level evidence on the dynamic relationship. The results suggest that the growth effects of currency appreciation and depreciation depend on a country's level of development and economic structure. Moderate exchange-rate movements may support growth, whereas excessive appreciation can weaken export competitiveness and excessive depreciation can increase inflation and reduce purchasing power. The findings underscore the importance of country-specific characteristics and exchange-rate regimes when designing policies to support sustainable economic growth.

Keywords: real exchange rate, economic growth, panel regression, time-series analysis

The Impact of the Real Exchange Rate on Economic Growth: A Comparative Study of OECD and Non-OECD Countries

Changes in the real exchange rate (RER) are an important channel through which globalization can affect a country's economic performance. In recent decades, economies such as Japan, South Korea, Singapore, and China have experienced rapid growth, which has often been associated with undervalued real exchange rates (Hu & Guo, 2021). However, the effects of RER movements on economic growth vary across economies; currency undervaluation does not necessarily promote growth in every context. Evidence from eastern and western China similarly suggests that the growth effects of RER movements differ across regions with different levels of economic development (Yang et al., 2023).

Existing studies mainly examine the effect of the RER on economic growth in individual countries or regions. Few studies, however, compare this relationship between OECD and non-OECD countries. This paper addresses that gap by examining both the differences between the two groups and the nonlinear nature of the RER-growth relationship.

For the purposes of this study, the sample is divided into OECD and non-OECD countries to capture broad differences in economic development, institutional structure, and exposure to external shocks. Most OECD members are advanced economies with established economic structures and policy frameworks, whereas many non-OECD members, particularly developing and emerging economies, are more vulnerable to external shocks. Accordingly, this paper uses economic reasoning and empirical analysis to compare the effects of RER movements on growth in the two groups, drawing on panel regressions and time-series evidence.

Gross domestic product (GDP) is widely used as an indicator of a country's economic performance and can be measured using several approaches. Under the expenditure approach, GDP is the sum of household consumption (C), business investment (I), government expenditure (G), and net exports ($X - M$). Under the production approach, GDP is measured by aggregating value added across sectors. The RER represents the price of domestic goods relative to foreign goods. In this study, it is constructed as the inverse of PWT's output-side GDP price level,

$RER_{i,t} = 1/pl_gdp_{i,t}$. Because PWT's price-level measure is purchasing power parity divided by the market exchange rate relative to the United States, a higher value of RER under this convention indicates a real depreciation, whereas a lower value indicates a real appreciation.

This paper uses fixed-effects panel regressions to examine the relationship between the RER and economic growth in OECD and non-OECD countries. Vector autoregression (VAR) models for China and Japan provide complementary country-level time-series evidence and are used to assess whether the panel findings are also observed in individual economies.

Literature Review

Existing research generally supports the view that real exchange-rate undervaluation can promote economic growth. Rodrik (2008) argues that currency undervaluation supports growth by expanding the tradable sector. Habib et al. (2017) find that RER undervaluation increases GDP growth, although mainly in developing countries and countries with pegged exchange rates. Other studies, however, identify possible adverse effects. Haddad and Pancaro (2010) argue that RER undervaluation may be unsustainable because its export- and growth-promoting effects diminish over time and may be accompanied by macroeconomic costs. Conrad and Jagessar (2018) find that RER misalignment negatively affects economic growth.

Recent studies also suggest that the relationship between the RER and economic performance is nonlinear. Missio et al. (2015) find that maintaining a moderately undervalued RER can support growth, particularly in Latin America, but that the effect is nonlinear. Yang et al. (2023) show that the growth-promoting effect of a lower real effective exchange rate through international trade varies with the exchange-rate level and differs across China's provinces. In related evidence from the East African Community, Nnyanzi et al. (2022) report an inverted U-shaped association between RER undervaluation and services-sector performance.

Modeling

Data Collection and Preprocessing

The Penn World Table (PWT), version 10.0, provides the raw data for this study (Feenstra et al., 2015, 2021). PWT contains information on relative levels of income, output, inputs, and

productivity for 183 countries from 1950 to 2019. It also provides alternative GDP measures and price-level data from which growth rates and the RER can be constructed. The analysis is restricted to 1970–2019, and observations with missing values for the model variables are excluded. Growth rates are calculated before the study-period restriction so that the 1970 values use the corresponding 1969 base observations.

The sample classification follows OECD membership during the 1970–2019 study window. Costa Rica, which joined the OECD in 2021, is therefore included in the non-OECD group. Complete-case filtering yields 37 OECD countries and 81 non-OECD countries. These rules reproduce the descriptive statistics and sample sizes reported in Tables 2 and 3 from the PWT 10.0 workbook.

Table 1 provides a preliminary summary of the study variables. The ranges indicate substantial dispersion and potential outlying observations, particularly in GDP, population, consumption, and growth. These observations were examined and retained when they were judged to be plausible.

Table 1
Descriptive Statistics for the Study Variables

Statistic	cgdpo (millions)	pop (millions)	ccon (millions)	growth	RER
Minimum	20	0.05	18	-0.539	0.05
First quartile	6,655	1.20	5,785	0.003	1.93
Median	31,330	6.54	25,127	0.041	2.87
Mean	3.62×10^5	33.78	2.64×10^5	0.045	3.64
Third quartile	1.99×10^5	21.66	1.37×10^5	0.081	4.44
Maximum	2.06×10^7	1,433.78	1.68×10^7	0.889	53.07

Note. cgdpo = output-side real gross domestic product at current purchasing power parities; pop = population; ccon = real consumption of households and government at current purchasing power parities; RER = real exchange rate. Monetary values are expressed in millions of 2017 U.S. dollars.

As expected for mature economies, OECD countries have a lower average growth rate than non-OECD countries in the study sample.

Methodology

Panel and VAR models are widely used in macroeconomics for forecasting, identifying turning points, and comparing the effects of monetary policy (Canova & Ciccarelli, 2004; Gerlach & Smets, 1995). In this study, the panel and VAR analyses are estimated separately so that they provide complementary evidence and help address the limitations of either model used alone.

Panel Data Analysis

Fixed-Effects Model

A fixed-effects model controls for unobserved characteristics that vary across entities but remain constant over time. It estimates within-country relationships by accounting for time-invariant country characteristics (Wooldridge, 2010). To control for unobserved heterogeneity, the sample is divided into OECD and non-OECD countries, and separate fixed-effects regressions are estimated for each group.

The fixed-effects model is specified as follows:

$$\text{GDPr}_{i,t} = \beta_0 + \beta_1 \text{RER}_{i,t} + \beta_2 \text{RER}_{i,t}^2 + \sum_{k=3}^7 \beta_k \text{Control}_{k,i,t} + \alpha_i + \lambda_t + \mu_{i,t}. \quad (1)$$

In Equation 1, $\text{GDPr}_{i,t}$ is the output-side real-GDP growth rate for country i in year t ; β_0 is the intercept; and $\beta_1, \dots, \beta_7$ are estimated coefficients. $\text{RER}_{i,t}$ is the real exchange rate, and $\text{RER}_{i,t}^2$ is its squared term. The control variables are $\ln\text{pop}$ (the natural logarithm of population, measured in millions), emp (the employment rate), $\ln\text{hc}$ (the natural logarithm of the human-capital index), $\ln\text{con}$ (the natural logarithm of real household and government consumption at current purchasing power parities, measured in millions of 2017 U.S. dollars), and labsh (labor compensation as a share of GDP at current national prices). Finally, α_i is the country fixed effect, λ_t is the year fixed effect, and $\mu_{i,t}$ is the idiosyncratic error term.

Descriptive Statistics

Data for OECD countries cover 1970–2019. Variables with large scales are log-transformed, and complete cases are retained. After preprocessing, the OECD sample contains 1,722 observations. Table 2 reports the descriptive statistics.

Table 2
Descriptive Statistics for OECD Countries

Variable	<i>n</i>	<i>M</i>	<i>SD</i>	Minimum	<i>Mdn</i>	Maximum
GDP _r	1,722	0.033	0.048	-0.296	0.033	0.308
GDP _{r2}	1,722	0.033	0.045	-0.287	0.034	0.398
RER	1,722	2.294	1.635	0.548	1.711	10.908
lnpop	1,722	2.557	1.517	-1.588	2.354	5.796
emp	1,722	0.444	0.070	0.234	0.449	0.748
lnhc	1,722	1.063	0.191	0.271	1.107	1.359
lncon	1,722	12.351	1.547	7.791	12.248	16.638
labsh	1,722	0.578	0.077	0.317	0.587	0.771

Note. GDP_r is the growth rate of output-side real GDP at chained purchasing power parities. GDP_{r2} is the growth rate of expenditure-side real GDP at chained purchasing power parities. RER = real exchange rate; lnpop = log population; emp = employment rate; lnhc = log human-capital index; lncon = log real consumption; labsh = labor compensation as a share of GDP.

Data for non-OECD countries also cover 1970–2019. After the same preprocessing steps, the non-OECD sample contains 3,726 observations. Table 3 reports the descriptive statistics. The count is independently reproduced from the supplied PWT 10.0 workbook using the model variables and the study-period OECD classification.

Table 3*Descriptive Statistics for Non-OECD Countries*

Variable	<i>n</i>	<i>M</i>	<i>SD</i>	Minimum	<i>Mdn</i>	Maximum
GDP _r	3,726	0.045	0.097	-0.799	0.045	0.994
GDP _{r2}	3,726	0.046	0.094	-0.813	0.045	1.089
RER	3,726	3.750	2.282	0.052	3.102	19.847
lnpop	3,726	2.118	1.675	-2.125	2.032	7.268
emp	3,726	0.375	0.086	0.176	0.371	0.763
lnhc	3,726	0.668	0.312	0.007	0.697	1.471
lncon	3,726	10.556	1.659	6.460	10.324	16.188
labsh	3,726	0.501	0.135	0.090	0.502	0.903

Note. GDP_r is the growth rate of output-side real GDP at chained purchasing power parities. GDP_{r2} is the growth rate of expenditure-side real GDP at chained purchasing power parities. RER = real exchange rate; lnpop = log population; emp = employment rate; lnhc = log human-capital index; lncon = log real consumption; labsh = labor compensation as a share of GDP.

Two chained-PPP GDP measures are used to construct the dependent variables: output-side real GDP (rgdpo) and expenditure-side real GDP (rgdpe). Both measures are converted to annual growth rates, yielding GDP_r and GDP_{r2}, respectively. To allow for a nonlinear relationship between the RER and GDP growth, the regressions include both RER and its squared term, RER². Control variables were selected based on theoretical relevance and data availability. Variables with large scales are log-transformed. To reduce collinearity between employment and population, the number of employed persons is converted to an employment rate.

Correlation Matrix

Pearson correlation coefficients were examined before the regression analysis. Tables 4 and 5 show that RER and RER² are significantly correlated with GDP_r, consistent with the hypothesized nonlinear relationship. Several controls, including emp and lnhc, are also significantly correlated with GDP_r. Because bivariate correlations do not account for other controls or for country and year effects, these results are descriptive; multivariate regressions are required to assess the conditional relationships.

Table 4
Correlation Matrix for OECD Countries

Variable	GDP _r	RER	RER2	lnpop	emp	lnhc	lncon	labsh
GDP _r	1.000							
RER	0.109***	1.000						
RER2	0.133***	0.947***	1.000					
lnpop	-0.004	0.080***	0.067***	1.000				
emp	-0.080***	-0.484***	-0.395***	-0.315***	1.000			
lnhc	-0.115***	-0.636***	-0.567***	-0.066***	0.634***	1.000		
lncon	-0.046*	-0.176***	-0.144***	0.942***	-0.090***	0.179***	1.000	
labsh	-0.103***	-0.084***	-0.084***	-0.108***	0.415***	0.316***	-0.025	1.000

Note. RER2 denotes RER^2 . * $p < .10$. ** $p < .05$. *** $p < .01$.

Table 5
Correlation Matrix for Non-OECD Countries

Variable	GDP _r	RER	RER2	lnpop	emp	lnhc	lncon	labsh
GDP _r	1.000							
RER	0.044***	1.000						
RER2	0.036**	0.938***	1.000					
lnpop	0.015	0.065***	0.021	1.000				
emp	-0.004	-0.184***	-0.140***	0.085***	1.000			
lnhc	-0.021	-0.302***	-0.227***	-0.073***	0.274***	1.000		
lncon	0.032*	-0.108***	-0.089***	0.839***	0.222***	0.327***	1.000	
labsh	-0.099***	0.097***	0.057***	0.058***	-0.013	-0.070***	-0.065***	1.000

Note. RER2 denotes RER^2 . * $p < .10$. ** $p < .05$. *** $p < .01$.

Hausman and F Tests

The model-selection tests reported in Tables 6 and 7 support the use of fixed-effects models. Both the Hausman and F tests reject their respective null hypotheses; fixed-effects specifications are therefore used.

Table 6
Hausman and F Tests for OECD Countries

Test	Statistic	p	Decision
Hausman test	$\chi^2 = 28.95$	$< .001$	Reject
F test	$F = 3.16$	$< .001$	Reject

Note. Rejection in both tests supports the fixed-effects specification reported in the manuscript.

Table 7*Hausman and F Tests for Non-OECD Countries*

Test	Statistic	<i>p</i>	Decision
Hausman test	$\chi^2 = 40.92$	$< .001$	Reject
F test	$F = 2.15$	$< .001$	Reject

Note. Rejection in both tests supports the fixed-effects specification reported in the manuscript.

Baseline Regression

To assess the stability of the results across specifications, control variables are added sequentially. Table 8 reports the baseline regressions for OECD countries. In Model 1, RER is not statistically significant. After RER2 is added in Model 2, RER2 is positive and significant at the 1% level, whereas RER is negative and significant at the 1% level. The estimates therefore indicate a statistically significant U-shaped relationship between RER and GDPr. This pattern remains in the fully adjusted specification.

Table 8*Baseline Regression Results for OECD Countries*

Variable	Model 1 GDP _r	Model 2 GDP _r	Model 3 GDP _r
RER	-0.0028 (-1.61)	-0.0215*** (-5.46)	-0.0229*** (-5.61)
RER2		0.0017*** (5.30)	0.0016*** (4.50)
lnpop			-0.0026 (-0.19)
emp			0.0401 (1.13)
lnhc			-0.0031 (-0.11)
lncon			-0.0274*** (-3.08)
labsh			-0.2047*** (-5.36)
Constant	0.0394*** (9.50)	0.0685*** (9.99)	0.5223*** (5.11)
Observations	1,722	1,722	1,722
R^2	0.299	0.311	0.327
Country fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes

Note. t statistics are in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$. RER2 denotes RER^2 .

Table 9 reports the corresponding results for non-OECD countries. In Model 2, RER2 is negative and significant at the 1% level, whereas RER is positive and significant at the 1% level, indicating a statistically significant inverted U-shaped relationship between RER and GDP_r. This pattern remains in the fully adjusted specification.

Table 9*Baseline Regression Results for Non-OECD Countries*

Variable	Model 1 GDP _{it}	Model 2 GDP _{it}	Model 3 GDP _{it}
RER	0.0028** (2.20)	0.0129*** (4.25)	0.0098*** (3.16)
RER2		-0.0007*** (-3.66)	-0.0005*** (-2.77)
lnpop			0.0004 (0.04)
emp			-0.1248*** (-2.84)
lnhc			-0.1182*** (-4.57)
lncon			0.0231*** (4.11)
labsh			-0.1356*** (-4.16)
Constant	0.0343*** (6.78)	0.0096 (1.14)	-0.0326 (-0.59)
Observations	3,726	3,726	3,726
R^2	0.137	0.140	0.153
Country fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes

Note. t statistics are in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$. RER2 denotes RER^2 .

Model Diagnostics and Robustness

Heterogeneity and Homogeneity Tests. To examine whether the relationship varies with population size, countries are divided into large- and small-population groups using the sample median as the cutoff. Table 10 reports the results for OECD countries.

Table 10*Heterogeneity by Population Size in OECD Countries*

Variable	Large population GDP _r	Small population GDP _r
RER	-0.0199*** (-4.07)	-0.0330*** (-4.47)
RER2	0.0015*** (3.84)	0.0025*** (3.31)
lnpop	0.0582*** (2.97)	-0.0491** (-2.43)
emp	-0.1094** (-2.01)	0.1856*** (3.65)
lnhc	-0.0232 (-0.60)	-0.0340 (-0.77)
lncon	-0.0210** (-2.14)	-0.0215 (-1.22)
labsh	-0.0267 (-0.46)	-0.3716*** (-6.96)
Constant	0.2179 (1.64)	0.5640*** (3.03)
Observations	828	894
R^2	0.369	0.382
Country fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Note. t statistics are in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$. Groups are defined using the sample median population. RER2 denotes RER^2 .

For OECD countries, the U-shaped relationship persists in both population groups. However, the coefficient magnitudes and implied turning points differ, indicating heterogeneity in the strength of the relationship. Table 11 reports the corresponding estimates for non-OECD countries. In this group, the inverted U-shaped relationship is statistically significant only among countries with smaller populations.

Table 11*Heterogeneity by Population Size in Non-OECD Countries*

Variable	Large population GDP _{Pr}	Small population GDP _{Pr}
RER	0.0032 (0.77)	0.0143*** (2.97)
RER2	0.0000 (0.02)	-0.0008*** (-2.93)
lnpop	-0.0037 (-0.21)	0.0053 (0.37)
emp	-0.1884*** (-2.61)	-0.0664 (-1.06)
lnhc	-0.1554*** (-4.27)	-0.0679* (-1.70)
lncon	0.0412*** (5.96)	-0.0044 (-0.45)
labsh	-0.0469 (-1.12)	-0.2462*** (-4.65)
Constant	-0.2470** (-2.46)	0.2450*** (2.97)
Observations	1,833	1,893
R^2	0.177	0.171
Country fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Note. t statistics are in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$. Groups are defined using the sample median population. RER2 denotes RER^2 .

Endogeneity: Two-Stage Least Squares. The source manuscript reports an exploratory two-stage least-squares (2SLS) specification that uses the first-order lag of RER as an instrument for current RER. Table 12 reports the results for OECD countries. In the reported first stage, lagged RER is significantly associated with current RER. The underidentification and weak-identification statistics are reproduced as reported, but they do not establish instrument exogeneity. Moreover, because the second stage includes both RER and RER2 while the source output provides only one first-stage equation, the available results do not fully document identification of the nonlinear specification.

Table 12*Two-Stage Least-Squares Results for OECD Countries*

Variable	First stage RER	2SLS GDP _r
L.RER	0.8257*** (75.73)	
RER		-0.0189*** (-4.04)
RER2		0.0017*** (4.18)
lnpop	0.0050 (0.07)	0.0015 (0.11)
emp	-0.0488 (-0.24)	0.0287 (0.81)
lnhc	-0.5452*** (-3.22)	0.0087 (0.29)
lncon	-0.1760*** (-3.43)	-0.0189** (-2.09)
labsh	-0.0351 (-0.16)	-0.1951*** (-5.17)
Constant	3.0742*** (5.21)	
Underidentification test		1,321.378***
Weak-identification test		2,894.432***
Observations	1,685	1,685
R^2	0.979	0.033
Country fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Note. t statistics are in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$. L.RER denotes the first-order lag of RER; RER2 denotes RER^2 . The source output reports one first-stage equation although the second stage contains RER and RER2; the nonlinear identification strategy is therefore incompletely documented, and the estimates should be treated as exploratory.

The reported second-stage estimates retain a significant U-shaped relationship between RER and GDP_r at the 1% level, consistent with the baseline results. Table 13 presents the corresponding analysis for non-OECD countries. The coefficient on RER2 remains negative, and the inverted U-shaped pattern is preserved. Given the incomplete first-stage documentation, however, these 2SLS results are treated as exploratory rather than definitive evidence that endogeneity has been resolved.

Table 13*Two-Stage Least-Squares Results for Non-OECD Countries*

Variable	First stage RER	2SLS GDP _r
L.RER	0.8049*** (111.25)	
RER2		-0.0030*** (-8.66)
RER		0.0432*** (8.63)
lnpop	0.2417*** (4.14)	-0.0080 (-0.75)
emp	-0.1431 (-0.57)	-0.0869* (-1.90)
lnhc	-0.1276 (-0.87)	-0.1124*** (-4.21)
lncon	0.0869*** (2.76)	0.0163*** (2.77)
labsh	-0.4692** (-2.56)	-0.1146*** (-3.42)
Constant	-0.4695 (-1.48)	
Underidentification test		1,617.310***
Weak-identification test		2,798.821***
Observations	3,645	3,645
R^2	0.943	-0.005
Country fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Note. t statistics are in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$. L.RER denotes the first-order lag of RER; RER2 denotes RER^2 . The 81-observation reduction from the baseline sample is consistent with using a first-order lag. The source output reports one first-stage equation although the second stage contains RER and RER2; the nonlinear identification strategy is therefore incompletely documented, and the estimates should be treated as exploratory.

Robustness Tests. To reduce dependence on the choice of dependent variable, the robustness test replaces GDP_r with the alternative measure GDP_{r2}. Table 14 reports the results for OECD countries, and Table 15 reports the results for non-OECD countries. The estimates using the alternative dependent variable are consistent with the baseline results, supporting the robustness of the findings.

Table 14*Robustness Tests for OECD Countries*

Variable	Model 1 GDPr2	Model 2 GDPr2
RER	-0.0240*** (-6.45)	-0.0252*** (-6.58)
RER2	0.0018*** (5.66)	0.0015*** (4.61)
lnpop		-0.0065 (-0.51)
emp		0.0249 (0.75)
lnhc		-0.0243 (-0.88)
lncon		-0.0270*** (-3.22)
labsh		-0.2076*** (-5.79)
Constant	0.0745*** (11.52)	0.5640*** (5.86)
Observations	1,722	1,722
R^2	0.305	0.326
Country fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Note. t statistics are in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$. GDPr2 is the growth rate of expenditure-side real GDP. RER2 denotes RER^2 .

Table 15*Robustness Tests for Non-OECD Countries*

Variable	Model 1 GDPr2	Model 2 GDPr2
RER	0.0097*** (3.31)	0.0068** (2.26)
RER2	-0.0005*** (-2.82)	-0.0004** (-1.97)
lnpop		-0.0069 (-0.70)
emp		-0.1776*** (-4.17)
lnhc		-0.1212*** (-4.84)
lncon		0.0240*** (4.42)
labsh		-0.1409*** (-4.46)
Constant	0.0189** (2.31)	0.0063 (0.12)
Observations	3,726	3,726
R^2	0.133	0.150
Country fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Note. t statistics are in parentheses. * $p < .10$. ** $p < .05$. *** $p < .01$. GDPr2 is the growth rate of expenditure-side real GDP. RER2 denotes RER^2 .

Panel Analysis Summary and Discussion

For OECD countries, the relationship between the RER and economic growth is U-shaped. At relatively low RER levels, an increase in the RER is associated with lower growth, whereas at relatively high RER levels, it is associated with higher growth. For non-OECD countries, the relationship is inverted U-shaped: an increase in the RER is associated with higher growth at relatively low RER levels and with lower growth at relatively high RER levels. Because a higher RER represents real depreciation under the study's variable construction, these patterns should be interpreted in terms of the level-dependent association between depreciation and growth rather than as a uniform benefit of either appreciation or depreciation.

Vector Autoregression Model

Vector autoregression is widely used in econometrics and time-series analysis to model dynamic relationships among multiple variables. In a VAR model, each variable depends on its own past values and on the past values of the other variables in the system. Theory-based restrictions can also be used to examine short- and long-run relationships among prices, output, and real exchange rates (Jacobson et al., 1999).

The VAR(p) model is specified as follows:

$$\mathbf{Y}_t = \mathbf{C} + \Phi_1 \mathbf{Y}_{t-1} + \cdots + \Phi_p \mathbf{Y}_{t-p} + B\mathbf{X}_t + \boldsymbol{\varepsilon}_t, \quad t = 1, 2, \dots, T. \quad (2)$$

In Equation 2, $\mathbf{Y}_t$ is a k -dimensional vector of endogenous variables; $\mathbf{Y}_{t-i}$, for $i = 1, \dots, p$, is the i th lag of $\mathbf{Y}_t$; $\mathbf{C}$ is a k -dimensional vector of constants; Φ_i is a coefficient matrix to be estimated; $\mathbf{X}_t$ is a vector of exogenous variables; and $\boldsymbol{\varepsilon}_t$ is a vector of error terms.

In this study, the VAR analysis complements the panel regressions. China and Japan are analyzed in detail to assess the panel findings at the country level, while other countries are discussed briefly. To reproduce the country models, growth is defined as the annual proportional change in output-side real GDP at current purchasing power parities (cgdp_o). This differs from the chained-PPP growth measure used as the main dependent variable in the panel regressions and should be considered when comparing the two analyses.

VAR Results

Stability and Granger-Causality Tests. Before estimating the VAR models, the stationarity of growth, RER, and RER2 is evaluated. The augmented Dickey-Fuller results in Table 16 reject the unit-root null hypothesis at the 5% level for all three series in China and Japan. The reported RER and RER2 statistics are reproduced exactly from the supplied workbook using one lag and no constant or trend; the corresponding growth tests use the same specification.

Table 16*Augmented Dickey-Fuller Tests*

Country	Series	Statistic	<i>p</i>	Decision
China	growth	-2.15	.030	Reject
China	RER	-2.87	.004	Reject
China	RER2	-4.09	< .001	Reject
Japan	growth	-2.32	.020	Reject
Japan	RER	-4.54	< .001	Reject
Japan	RER2	-10.03	< .001	Reject

Note. The null hypothesis is that the series contains a unit root. Tests use one lag and include neither a constant nor a deterministic trend. Growth is the annual proportional change in *cgdpo*. RER2 denotes RER^2 .

Granger-causality tests are also conducted after the lag lengths are selected using the Akaike information criterion. Table 17 shows that the joint null hypothesis that growth does not Granger-cause RER and RER2 is rejected for both China and Japan. The joint null hypothesis for RER as a predictor of growth and RER2 is not rejected in either country, whereas the joint null hypothesis for RER2 as a predictor of growth and RER is rejected for China but not for Japan. The direction of the predictive relationship therefore differs across the two countries. These results concern predictive precedence and should not be interpreted as structural causation.

Table 17*Granger-Causality Tests*

Country	Null hypothesis	<i>p</i>	Decision
China	Growth does not Granger-cause RER and RER2	.0094	Reject
China	RER does not Granger-cause growth and RER2	.065	Fail to reject
China	RER2 does not Granger-cause growth and RER	.020	Reject
Japan	Growth does not Granger-cause RER and RER2	.0055	Reject
Japan	RER does not Granger-cause growth and RER2	.170	Fail to reject
Japan	RER2 does not Granger-cause growth and RER	.063	Fail to reject

Note. Each row reports a joint test for the two outcomes named in the null hypothesis. Lag lengths were selected using the Akaike information criterion. Granger causality indicates predictive precedence, not structural causation. RER2 denotes RER^2 .

Coefficients and Impulse Responses. For China, the Akaike information criterion selects a VAR(2) model. Table 18 reports the growth-equation coefficients; the coefficient estimates are independently reproduced from the supplied workbook. Their signs are broadly

consistent with the panel regression that includes a first-order lag and suggest an inverted U-shaped dynamic relationship between the RER and economic growth. This pattern suggests that increases in the RER may support growth up to a threshold, beyond which further increases are associated with lower growth.

Table 18

Growth-Equation Estimates From the VAR(2) Model for China

Coefficient	Estimate	SE	p
growth.l1	0.286	0.140	.047*
RER.l1	0.133	0.045	.005**
RER2.l1	-0.008	0.003	.007**
growth.l2	0.028	0.142	.844
RER.l2	-0.114	0.044	.013*
RER2.l2	0.006	0.003	.026*
Constant	0.020	0.025	.427

Note. The dependent variable is growth, defined as the annual proportional change in cgdpo. l1 and l2 denote first- and second-order lags. RER2 denotes RER^2 . * $p < .05$. ** $p < .01$.

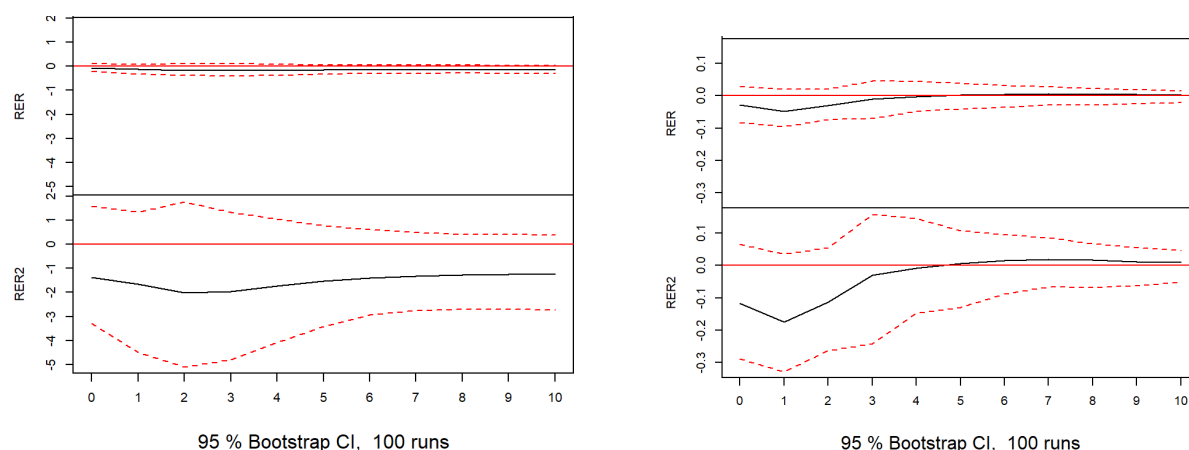
One root of the characteristic equation slightly exceeds unity, so the China model does not fully satisfy the stability condition. The model is nevertheless retained for exploratory analysis of the dynamic relationships, and its results should be interpreted cautiously.

For Japan, the Akaike information criterion selects a VAR(3) model. Table 19 reports the subset of growth-equation coefficients available in the source manuscript; the coefficient estimates are independently reproduced from the supplied workbook. The first-lag coefficients are consistent with the U-shaped relationship found in the panel analysis, although the evidence is less pronounced. All roots of the Japan model are less than one, so the model satisfies the stability condition and is dynamically stable.

Table 19*Partial Growth-Equation Estimates From the VAR(3) Model for Japan*

Coefficient	Estimate	SE	p
growth.l1	0.237	0.148	.118
RER.l1	-0.177	0.067	.012*
RER2.l1	0.035	0.014	.019*
growth.l2	0.240	0.148	.113
RER.l2	0.034	0.075	.650
RER2.l2	-0.009	0.013	.480
...			

Note. The dependent variable is growth, defined as the annual proportional change in cgdpo. The source manuscript reports only part of the VAR(3) growth equation. l1 and l2 denote first- and second-order lags. RER2 denotes RER^2 . * $p < .05$.

Figure 1*Responses of RER and RER2 to an Economic-Growth Shock in China and Japan***(a)** Responses of RER and RER2 to an Economic-Growth Shock in China**(b)** Responses of RER and RER2 to an Economic-Growth Shock in Japan

Note. Solid lines show the estimated impulse responses; dashed lines show 95% bootstrap confidence intervals based on 100 runs. RER2 denotes RER^2 .

Figure 1a shows that, following a positive shock to economic growth in China, the response of RER is small and remains near zero, whereas RER2 responds negatively and gradually converges toward zero. The estimates therefore describe the adjustment of the exchange-rate variables to a growth shock, not the effect of an RER shock on economic growth. Because the China VAR does not fully satisfy the stability condition, the response should be treated as exploratory.

Figure 1b shows that, following a positive shock to economic growth in Japan, RER initially declines and then returns toward zero. RER2 also declines initially, then rises toward zero and becomes slightly positive in later periods before converging. These patterns indicate a transitory adjustment of the exchange-rate variables to a growth shock in Japan.

VAR Summary and Discussion

Although the panel and country-level analyses generally indicate U-shaped and inverted U-shaped relationships, some countries depart from the group-level patterns. For example, the signs of the coefficients in the VAR model for the United States resemble those for China, even though the United States is an OECD member. The omission of additional control variables from the VAR models may help explain why some country-level results differ from the panel estimates and from previous studies (Jacobson et al., 1999).

Discussion and Limitations

Using panel regression and time-series analyses, this study finds that the RER is significantly associated with economic growth and that the estimated relationship differs between OECD and non-OECD countries. The panel results indicate a U-shaped relationship for OECD countries and an inverted U-shaped relationship for non-OECD countries. The country-level VAR analyses provide complementary evidence but also show that individual economies can depart from the group-level patterns.

The construction of RER is important for interpreting these findings. Because RER is the inverse of PWT's output-side GDP price level, an increase represents real depreciation and a decrease represents real appreciation. Real depreciation can improve export price competitiveness, but it can also raise the domestic cost of imported goods and intensify inflationary pressure. Real appreciation can lower import costs and support purchasing power, but it may weaken export competitiveness. Theoretical and empirical work therefore emphasizes that the growth consequences of exchange-rate movements depend on economic structure, the tradable sector, and the initial degree of misalignment (Edwards, 1989; Rodrik, 2008).

For OECD countries, the estimated U shape means that the marginal association between

further depreciation and growth changes sign across the observed RER range. This result does not imply that either appreciation or depreciation is always desirable. Models of consumption and real exchange rates also show that incomplete international asset markets and nontraded goods can materially affect the adjustment process (Benigno & Thoenissen, 2008). Export-oriented OECD economies may benefit from greater price competitiveness after depreciation, but higher import costs and inflation can offset those benefits.

For non-OECD countries, the inverted U shape likewise indicates that real depreciation is associated with higher growth only up to an estimated threshold. Beyond that point, further depreciation is associated with lower growth. This pattern is consistent with the trade-off emphasized in studies of developing economies: depreciation may support exports and adjustment, while high import costs, inflation, and foreign-currency liabilities can constrain activity (Edwards, 1989; Haddad & Pancaro, 2010).

Taken together, the results indicate that both groups must balance exchange-rate objectives with their economic structures. OECD countries, with more complex industrial and financial systems, should emphasize innovation and technological upgrading rather than rely on persistent exchange-rate changes. For non-OECD countries, economic diversification, infrastructure investment, and industrial upgrading may reduce dependence on a narrow export base. These countries must also manage inflation and capital flows while adjusting exchange-rate policy to domestic and external conditions.

Exchange-rate regimes introduce an additional policy dimension that is not modeled directly in this study. Fixed regimes can reduce nominal exchange-rate variation but constrain monetary-policy adjustment, whereas floating regimes provide greater flexibility but can expose less-developed financial systems to volatility. Research on regime choice and performance shows that outcomes differ across advanced, emerging, and developing economies (Ghosh et al., 2003; Rogoff et al., 2003). Evidence on currency crashes in emerging markets further underscores the importance of reserve adequacy, external conditions, and domestic vulnerability when evaluating regime sustainability (Frankel & Rose, 1996). The present estimates should therefore not be

interpreted as recommending a single exchange-rate regime for either group.

Several limitations remain. First, the quadratic specification summarizes the RER-growth relationship as U-shaped or inverted U-shaped, but it cannot capture the full complexity of individual economies. Second, although the OECD/non-OECD classification provides a useful comparative framework, it may obscure differences in economic structure, political stability, and institutions. Third, the analysis does not explicitly model exchange-rate regimes. Fourth, the available 2SLS output does not fully document identification of both the linear and squared RER terms. Fifth, the China VAR has a root slightly above unity, so its dynamic results require caution. Finally, Table 19 contains only the subset of the Japan VAR(3) output available in the source manuscript. The estimated nonlinear relationships should therefore not be assumed to apply uniformly to every country.

Overall, the empirical results identify broad patterns in the relationship between the RER and economic growth, but they also show the importance of exchange-rate regimes and country-specific conditions. Effective policy must balance financial stability, external competitiveness, purchasing power, and sustainable growth.

Conclusion

This study examines the relationship between the RER and real GDP growth in OECD and non-OECD countries using fixed-effects panel regressions and country-level VAR analyses. The principal finding is a U-shaped relationship in the OECD sample and an inverted U-shaped relationship in the non-OECD sample. For OECD countries, the estimated marginal association between the RER and growth is negative at relatively low RER levels and positive at relatively high levels. For non-OECD countries, the corresponding association is positive at relatively low RER levels and negative at relatively high levels. These patterns are supported by the baseline, heterogeneity, and alternative-dependent-variable robustness analyses; the 2SLS results are directionally consistent but exploratory because the available first-stage documentation is incomplete.

The findings should nevertheless be interpreted cautiously. The OECD/non-OECD

classification may oversimplify substantial within-group differences, and the empirical models do not explicitly account for exchange-rate regimes or other country-specific institutional and political factors. The country-level VAR results also show that individual economies can depart from the group-level pattern, and the instability of the China VAR limits strong dynamic inference.

Despite these limitations, the results provide useful guidance for policymakers. OECD countries must balance export competitiveness against import costs and inflation, whereas non-OECD countries should combine export-oriented policies with economic diversification and inflation control. More generally, exchange-rate policy should be adapted to national circumstances rather than applied uniformly across countries. Such an approach can help policymakers manage exchange-rate trade-offs while supporting sustainable economic growth.

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